# Muhammad Umer

Machine Learning Engineer — Computer Vision, NLP, Reinforcement Learning & AI Agents

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## RESEARCH SUMMARY

Computer Science undergraduate with hands-on research experience in computer vision, temporal data, diffusion models, multimodal deep learning. First author of an IEEE-conference-targeted paper on CSR-MMDiff, a clear-sky-residual multimodal diffusion model for probabilistic solar PV forecasting. Builds LLM agent systems (LangGraph, RAG, tool-use) and has designed high-throughput data pipelines for large-scale multimodal datasets. Also focusing on Computer Vision. Seeking a PhD position to understand state-of-the-art Artificial Intelligence that is Deep Reinforcement Learning, AI Agents, and Computer Vision.

## EDUCATION

**BS Computer Science — University of Management and Technology (UMT), Lahore, Pakistan**
*Oct 2023 – Expected Jul 2027 | CGPA: 3.27/4.0 (6 semesters completed)*

## RESEARCH EXPERIENCE & PUBLICATIONS

**CSR-MMDiff: Clear-Sky-Residual Multimodal Diffusion for Probabilistic PV Forecasting**
*Research Project (targeting IEEE conference publication) | Python, TensorFlow, Diffusion Models (DDIM), U-Net, Cross-Modal Attention, pvlib*

- Proposed a joint diffusion model of future sky images and PV power on the Stanford SKIPP'D benchmark that generates only the residual from a clear-sky smart-persistence prior and the last sky frame, for 2–16-minute probabilistic forecasts.
- Designed a 5.1M-parameter coupled U-Net with cloud-aware cross-modal attention (learned spatial pooling of sky-image features), clear-sky physics conditioning, v-prediction and Min-SNR loss weighting.
- Built a projected DDIM sampler that enforces physical bounds at every step, plus lead-wise spread calibration fitted on validation days; 30-member, 20-step ensembles take 54 ms per member.
- Lowered CRPS to 0.943 kW (28.7% skill over smart persistence, 22.9% below our prior DDIM model) and raised 90% interval coverage from 77.9% to 83.2% with narrower intervals; generated frames keep SSIM 0.724 at +16 min (vs. 0.423).
- Trained with data-parallel mixed precision on two RTX 3080 GPUs (7.9 h) on day-blocked splits (99k/11k windows), with modality dropout so one model serves image-only, PV-only and joint inputs.

*Publication:* Muhammad Umer, Haseeb Khan, Jameel Ahmad, Syed Baqar Hussain, Abdul Hameed, "CSR-MMDiff: Clear-Sky-Residual Multimodal Diffusion for Probabilistic PV Forecasting," Dept. of Computer Science, UMT — IEEE Conference Paper (under preparation/submission).

## MACHINE LEARNING PROJECTS

**UrbanTrafficOptimization — Deep Learning & Computer Vision (PyTorch)**

- Combined an LSTM traffic-flow predictor with YOLOv8 vehicle detection to optimize signal/traffic control from camera imagery.

**SkinCancerImagesDetector & MiniDeepLearningProjects**

- Built a benign/malignant skin-cancer classifier comparing four Scikit-learn models, and a suite of TensorFlow projects spanning ANN, CNN, RNN/LSTM, deep and bidirectional RNN architectures.

## SOFTWARE ENGINEERING PROJECTS

Full-stack/DevOps projects spanning microservices, real-time systems, and applied Gen-AI, including **Ticketing** (Node.js microservices), **EIDOS** (AI-generated study kits, Next.js), **Greenly** (carbon-footprint tracker), **BooksExchange**, **BookIT**, and **DevBlogCms** (Next.js). Repositories: github.com/umerkang66.

## SKILLS

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| **AI / ML / RL** | Python, TensorFlow, Keras, PyTorch, Scikit-learn, Diffusion Models (DDIM), Deep Reinforcement Learning, Computer Vision, NLP, Multimodal Fusion, Data Analysis, Feature Engineering, Model Deployment |

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| **Agentic AI** | LangChain, LangChain JS, LangGraph, RAG, OpenAI API, Gemini API |
| **Data & Pipelines** | Pandas, NumPy (mmap), tf.data, PostgreSQL, MongoDB, Prisma, Drizzle, Database Design |
| **Software Eng. / DevOps** | C++, OOP, DSA, JavaScript/TypeScript, React & ecosystem, Next.js, Node.js/Express, Docker, Kubernetes, Linux, Git, NGINX, AWS/GCP/DigitalOcean/Linode |

## LANGUAGES

Urdu (native), English (proficient — academic and professional working)